IILM UNIVERSITY, GREATER NOIDA, U.P.

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

INTERNSHIP REPORT

ON

PYTHON PROGRAMMING VIRTUAL INTERNSHIP

Submitted in partial fulfilment of the requirement for the degree of

B.Tech CSE (AIML)

Submitted By

TANMAY GIRI

Roll / Enrollment No.: 25SCS1003000692

B.Tech CSE (AIML), Semester 3

Batch: 2026–27

Internship Organization: CodSoft

Internship Period: 01 August 2026 – 31 August 2026

CANDIDATE'S DECLARATION

I, TANMAY GIRI, hereby declare that the internship report titled "Python Programming Virtual Internship" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The work presented in this report is based on the tasks completed during the CodSoft Python Programming Internship. I affirm that the report is my own work and has not been submitted to any other institute for any degree or diploma requirement. I shall be solely responsible for any copyright violation arising from this report.

Signature: ___________________________

Student Name: TANMAY GIRI

Roll / Enrollment No.: 25SCS1003000692

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to everyone who supported and guided me throughout the internship program. I am thankful to the faculty members of the School of Computer Science and Engineering, IILM University, for their guidance and encouragement. I also acknowledge CodSoft for providing a project-based Python Programming Internship that allowed me to apply programming concepts through practical tasks.

I am grateful for the constructive feedback, learning opportunities and practical exposure received during the completion of the internship tasks. I also express my gratitude to my parents for their blessings and continuous support.

Signature: ___________________________

Student Name: TANMAY GIRI

Roll / Enrollment No.: 25SCS1003000692

INTERNSHIP COMPLETION CERTIFICATE

C.ID: 60f33ff

CodSoft

CERTIFICATE

OF COMPLETION
PROUDLY PRESENTED TO

Tanmay Giri

has successfully completed 4 weeks of a virtual internship program in
Python Programming
with wonderful remarks at **CODSOFT** from 01/08/2026 to 31/08/2026.

We were truly amazed by his/her showcased skills and invaluable contributions to the tasks and projects throughout the internship.

Founder

CERTIFIED ISO 9001-2015 COMPANY

MSME
MICRO, SMALL & MEDIUM ENTERPRISES

contact@codsoft.in www.codsoft.in Date: 01/09/2026

CodSoft certificate issued to Tanmay Giri for successful completion of the Python Programming virtual internship, dated 01/09/2026.

CodSoft

INTERNSHIP OFFER LETTER

Date : 30/07/2026

ID: BY26RY226424

Dear **Tanmay Giri,**

We would like to congratulate you on being selected for the **“Python Programming”** virtual internship position with **“CodSoft”**. We at CodSoft are excited that you will join our team. The duration of the internship will be of **1 Month**, starting from **01 August 2026 to 31 August 2026**. The internship is an educational opportunity for you hence the primary focus is on learning and developing new skills and gaining hands-on knowledge. We believe that you will perform all your tasks/projects. As an intern, we expect you to perform all assigned tasks to the best of your ability and follow any lawful and reasonable instructions provided to you.

We are confident that this virtual internship will be a valuable experience for you, we look forward to working with you and helping you achieve your career goals.
By accepting this offer, you commit to executing assigned tasks diligently and ensuring excellence in all aspects of your work.
Best of Luck!

ISO 9001 2015 CERTIFIED

ISO Number - TSNUK79053

Sincerely

Ruman Arshad
Program Manager

MSME
MICRO, SMALL & MEDIUM ENTERPRISES
सूक्ष्म, लघु एवं मध्यम उद्यम

MSME Regd

CODSOFT IT SERVICE CodSoft

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CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

This report presents the work completed during the CodSoft Python Programming Internship. The internship was designed around practical, project-based console applications that reinforce programming fundamentals, logic building and problem-solving skills. During the internship, three Python projects were completed: a Password Generator, a Rock-Paper-Scissors Game and a Contact Book.

The projects progressively apply user input, randomisation, conditional logic, loops, functions, collections and menu-driven interaction. The work demonstrates the use of Python to convert simple problem statements into functional command-line applications.

4.2 Organization Profile

CodSoft is described in the internship material as a technology education and software development platform offering project-based virtual internships. The Python Programming Internship focuses on building real-world console applications to strengthen core programming, logic-building and problem-solving skills.

The internship structure is practical in nature: each task has a defined objective, a set of functional features and an implementation approach. This enabled the intern to practise Python concepts through small, manageable applications.

4.3 Problem Statement

The project work addresses the need for practical application of basic Python programming concepts rather than learning them only through theory. The tasks require the development of simple console-based solutions for password generation, interactive game logic and contact management.

The main challenge is to design applications that accept user input, process it using appropriate Python constructs, produce correct results and provide a clear user experience through the console.

4.4 Project Objectives

- To strengthen fundamental Python programming and problem-solving skills.
- To practise console-based input and output and user interaction.
- To implement randomisation using Python’s random-related functionality.
- To apply conditional statements and loops to control program flow.
- To use lists and dictionaries for storing and managing application data.
- To develop functions for modular operations such as add, search, update and delete.
- To build clean, understandable and user-friendly command-line applications.

4.5 Scope of the Project

The scope of this internship work is limited to three console-based Python applications. The Password Generator creates passwords of a user-defined length. The Rock-Paper-Scissors Game provides an interactive user-versus-computer experience with result evaluation and optional score tracking. The Contact Book manages contact information through add, view, search, update and delete operations.

The applications are educational, lightweight and focused on Python fundamentals. The provided task material does not describe a graphical user interface, database server, cloud deployment or network-based functionality; therefore, these are outside the demonstrated scope of this report.

4.6 Technologies and Tools Used

Technology / Tool	Use in Project	Related Concepts
Python	Implementation of all three console applications	Variables, input/output, loops, conditions, functions
string module	Character pool for password generation	ascii_letters, digits, punctuation
random.choice()	Random password characters and computer game moves	Random selection
Lists / Dictionaries	Storage of contact information	Collection handling
Console I/O	Interaction with the user	Input and output
GitHub	Repository referenced in the internship submission	Code/project submission

Table 4.1: Technologies and tools used

4.7 System Architecture

All three applications follow a simple console-oriented processing architecture. The user provides input through the command line; the Python program validates or processes the input; the core logic performs the required operation; and the result is displayed back to the user. Repetition is controlled through loops where required.

User Input	Input Handling	Core Logic	Data / Random Source	Console Output
Choices / values	Validation & parsing	Task-specific processing	random / lists / dictionaries	Result / status / score

Figure 4.1: Common console application architecture used across the internship tasks

4.8 Methodology

- Requirement understanding: Identify the expected behaviour and features for each assigned task.

- Design: Break the task into input, processing, output and repetition/exit components.
- Implementation: Write Python code using suitable built-in modules, control structures and data structures.
- Validation: Check user input and ensure the program follows the required rules and operations.
- Execution and testing: Run the console application using sample inputs and observe the generated output.
- Documentation: Record the features, tools, approach and sample outputs for the internship report.

4.9 Expected Outcomes

- Improved understanding of Python programming fundamentals.
- Better ability to translate a problem statement into a working program.
- Practical experience with loops, conditions, functions and collections.
- Improved confidence in designing interactive console applications.
- Experience in presenting project features, implementation approach and sample outputs in a technical report.

4.10 Certificate and Communication Proof

The prescribed report format requires a copy of the internship completion certificate and relevant communication mail proof to be attached. The CodSoft completion certificate has been included in this report. Relevant communication proof, if required by the university, should be appended to the final submitted report.

4.11 Task 1 – Password Generator

The Password Generator is a command-line application that creates strong, random passwords based on a user-specified length. It combines uppercase and lowercase letters, digits and special symbols to form the character pool used for password generation.

4.11.1 Key Features

- Accepts the desired password length as user input.
- Combines uppercase letters, lowercase letters, digits and special symbols.
- Uses Python random / secrets functionality for random character selection.
- Prints the generated password directly to the console.
- Allows the user to generate another password.

4.11.2 Tools and Concepts Used

- Python string module: ascii_letters, digits and punctuation.
- random.choice() for random character selection.
- Loops and string concatenation.
- Input validation for password length.
- Console-based input/output.

4.11.3 Implementation Approach

Step	Activity	Purpose
1	Take Input	Ask the user to enter the desired password length.
2	Build Character Pool	Combine letters, digits and symbols into one character pool.
3	Generate Password	Randomly pick characters from the pool until the required length is reached.
4	Display Output	Print the generated password and optionally ask whether another password should be generated.

Table 4.2: Password Generator implementation steps

4.11.4 Sample Output

```
Enter the desired password length: 12
Generated Password: xQ7!vB2#mZk9
Do you want to generate another password? (y/n): n
```

Figure 4.2: Sample console output for Password Generator

4.12 Task 2 – Rock-Paper-Scissors Game

The Rock-Paper-Scissors Game is an interactive console application in which the user competes against the computer. The computer makes a random selection in each round, while conditional logic compares the two choices according to the standard game rules.

4.12.1 Key Features

- User chooses Rock, Paper or Scissors.
- Computer makes a random selection in each round.
- Game logic determines the winner using standard rules.
- The result of each round is displayed clearly.
- Score can be tracked across multiple rounds.
- A Play Again option allows the user to continue or exit.

4.12.2 Tools and Concepts Used

- random.choice() for the computer’s move.
- if / elif / else conditional statements for winner evaluation.
- while loop for repeated rounds.
- Variables for score tracking.
- Console-based user interface.

4.12.3 Implementation Approach

- User Input: User selects rock, paper or scissors.
- Computer Move: The system randomly picks its own move.
- Decide Winner: The two choices are compared using the game rules.
- Show Result & Score: The outcome and updated score are displayed and the user is asked whether to replay.

4.12.4 Sample Output

```
Choose rock, paper, or scissors: paper
Computer chose: rock
You win! Paper covers Rock.
Score -> You: 1 | Computer: 0
Play again? (y/n): n
```

Figure 4.3: Sample console output for Rock-Paper-Scissors Game

4.13 Task 3 – Contact Book

The Contact Book is a menu-driven console application designed to store and manage personal contact information. It supports create, read, update, delete and search operations and stores details such as name, phone number, email and address.

4.13.1 Key Features

- Stores name, phone number, email and address.
- Adds new contacts with complete details.
- Displays a list of saved contacts.
- Searches contacts by name or phone number.
- Updates existing contact details.
- Deletes a selected contact.

4.13.2 Tools and Concepts Used

- Python dictionaries / lists for data storage.
- Functions for add, view, search, update and delete operations.
- Menu-driven loop for user interaction.
- String matching for search functionality.
- Clean and user-friendly console interface.

4.13.3 Implementation Approach

Step	Operation	Description
1	Show Menu	Display Add, View, Search, Update, Delete and Exit options.
2	Take Choice	User selects an operation from the menu.

3	Perform Operation	The corresponding function runs on the contact list.
4	Loop / Exit	Return to the menu until the user chooses to exit.

Table 4.3: Contact Book implementation steps

4.13.4 Sample Output

```
1. Add Contact  2. View Contacts  3. Search  4. Update  5. Delete  6. Exit
Enter your choice: 1
Name: Rohan Sharma | Phone: 9876543210
Contact added successfully!
```

Figure 4.4: Sample console output for Contact Book

4.14 Conclusion

The CodSoft Python Programming Internship provided practical exposure to building small but functional console applications. Three projects were completed successfully: Password Generator, Rock-Paper-Scissors Game and Contact Book. Together, these tasks covered important Python fundamentals including user input, console output, random selection, conditional statements, loops, functions, lists and dictionaries.

The internship strengthened logic-building and problem-solving skills by requiring each application to be divided into clear steps such as input handling, core processing and output. The completed work also provided experience in designing user-friendly command-line interactions and documenting implementation approaches and sample outputs.

Overall, the internship work helped bridge the gap between theoretical Python concepts and their practical implementation through focused, real-world style programming tasks.

5. BIBLIOGRAPHY / REFERENCES

- CodSoft, “Python Programming Virtual Internship – Task Submission Report,” source material supplied for this report.
- Python Software Foundation, “Python Standard Library – random module,” reference for random selection used in the Password Generator and Rock-Paper-Scissors projects.
- Python Software Foundation, “Python Standard Library – string module,” reference for character constants used in the Password Generator.
- Python Software Foundation, “Python Tutorial – Data Structures,” reference for lists and dictionaries used in the Contact Book.
- Python Software Foundation, “Python Documentation – Built-in Types,” reference for Python strings and basic string operations used in the projects.
- GitHub, “CODSOFT_TASKNO,” repository referenced for the internship project submission.
- IILM University, “Internship Report Format updated 26-27,” report-format source document supplied for this report.